

Four Months' Notice

Turning El Niño into a dengue early-warning alarm for ASEAN

A freely available climate index gives ASEAN health offices months of warning before a dengue season turns severe. We turn that lag into a two-condition alert rule any provincial health office can run today — no model, no code, no paid data.

SDG 3 Health

SDG 13 Climate

SDG 6 Water

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THE PROBLEM

Dengue is ASEAN's fastest-moving climate-sensitive disease

1.26M

cases in 2019 — the worst year in our nine-country panel

- Vietnam 368k · Philippines 220k · Thailand 159k (latest complete year per country)
- The 2023–24 El Niño added an estimated **9.6 million cases worldwide** (Tian et al., 2025)
- Outbreaks are still fought **reactively** — after cases surge

ASEAN dengue cases, 2000–2024

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Line chart — *Dengue national annual*

Filter complete=True, year 2000–2024 · Measure cases (Sum)
· Dim X year · Reference line di 2019, label “record in our data”

gambar-sac/h02-tren-tahunan.png

Cases by country, latest complete year

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Bar chart horizontal — *Dengue national annual*

Measure cases · Dim country · filter tahun lengkap terbaru (2022–2023) ·urut menurun, biru; Indonesia disorot oranye

gambar-sac/h02-per-negara.png

THE QUESTION

Climate data arrives months before patients do

Can a freely available climate index tell a health office how bad the coming season will be — early enough to act?

- **SDG 3** Health — reduce dengue burden
- **SDG 13** Climate — use climate information for adaptation
- **SDG 6** Water — target the most vulnerable districts

Our test: does the signal survive when we remove the seasonal cycle, hold out unseen years, and use the WHO outbreak definition?

From public index to public action

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Diagram — SAC Shapes + Text (tanpa dataset)

Empat kotak berurutan, panah di antaranya: **Climate index (ONI, monthly, public)** → **3–4 month lag** → **Dengue surge** → **Action window: PSN/3M, larviciding, stockpiles**. Warna kotak mengikuti palet cerita: biru, abu, merah, oranye.

gambar-sac/h03-alur.png

DATA & METHOD

Nine open datasets, one monthly panel, zero paid sources

DATA	ROWS	COVERAGE
Dengue cases (OpenDengue V1.3)	70,397	10 countries, 1955–2025
Dengue × climate monthly panel	55,329	province-month, lags pre-computed
Temperature & rainfall (CCKP, ERA5)	9,000	monthly, 1950–2024
ONI El Niño index (NOAA CPC)	918	monthly, 1950–2026
Water & sanitation (WHO/UNICEF JMP)	8,692	2000–2024
Population (BPS Indonesia)	140	provinces, denominators
Flood exposure (GHSL)	30,348	836 cities, 1975–2030

SDG 3

SDG 13

SDG 6

Every table joins on the same two keys — iso3 + year (plus period for the monthly dengue × climate panel). Nothing is entered by hand.

METHOD

- Monthly national series, 2010+
- Lagged correlations, 0–4 months
- **Anomalies** — seasonal cycle removed before we believe any correlation
- A two-condition trigger rule
- Out-of-sample test on unseen years
- WHO/PAHO endemic-channel benchmark

All charts built in SAP Analytics Cloud.

ANALYSIS 1 — THE SEASON

Dengue has a calendar — and it is not the same calendar everywhere

- **Indonesia:** cases peak **Jan–Apr** (Jan avg 15,099/month vs Sep 6,003 — $2.5\times$), right after the December rainfall peak
- **Thailand:** cases climb with the **onset** of the wet season and peak **Jun–Sep** (Jul avg 8,110 vs Feb 1,465 — $5.5\times$)
- Same disease, opposite halves of the year → a regional alarm must be **local in timing**

Indonesia — cases vs rainfall by month

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Combination chart (Column & Line) — *Seasonality IDN-THA*

Filter country=Indonesia · Column (kiri): mean_cases, biru · Line (kanan, sumbu sekunder): mean_rain_mm, abu putus-putus · Dim X month (Sort by month) · 4 kolom tertinggi merah

gambar-sac/h05-musiman-idn.png

Thailand — cases vs rainfall by month

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Combination chart (Column & Line) — *Seasonality IDN-THA*

Setelan identik, filter country=Thailand. Catatan: puncak hujan Thailand justru September, *sesudah* puncak kasus (Juli) — karena itu narasinya “onset”, bukan “peak”.

gambar-sac/h05-musiman-tha.png

ANALYSIS 2 — THE LAG

The climate signal arrives 1–4 months before the cases

COUNTRY	BEST PREDICTOR	LAG	R
Indonesia	ONI (El Niño)	4 mo	+0.59
Indonesia	Temperature	3 mo	+0.47
Indonesia	Rainfall	1 mo	+0.40
Thailand	Temperature	3 mo	+0.57
Thailand	Rainfall	same mo	+0.48

The lag *is* the opportunity: it is the time a health office has to act.

Correlation by predictor and lag

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Heat map — Lag correlation - national

Filter iso3=IDN · Dim baris predictor · Dim kolom lag · Measure r · skala diverging -0,7... +0,7, nilai ditampilkan di sel. Widget kedua untuk THA kalau ruang cukup.

gambar-sac/h06-heatmap-jeda.png

Temperature (lag 2) vs cases, Indonesia

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Scatterplot — Monthly series IDN-THA

Filter country=Indonesia · X temp_c_lag2 · Y cases · warna titik enso_phase_lag4 · Trend line → Linear

gambar-sac/h06-scatter-suhu.png

ANALYSIS 3 — THE ACE CARD

A strong El Niño nearly triples Indonesia's monthly dengue burden

2.8×

21,383 vs 7,597 cases per month, grouped by the ENSO phase four months earlier

- Grouped by the phase **four months earlier** — so this is a forecast, not hindsight
- **Not just seasonality**: after removing the seasonal cycle the ONI correlation holds at **$r = +0.56$**
- **Not an aggregation artefact**: cases are higher in El Niño months in **33 of 34 Indonesian provinces** (sign test $p = 2 \times 10^{-9}$) and 70 of 77 Thai provinces
- Rainfall does *not* survive the same test ($+0.40 \rightarrow +0.24$) — we report that honestly

Mean monthly cases by ENSO phase (lag 4)

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Bar chart — ENSO phase - averages

Measure mean_cases_per_month · Dim enso_phase_lag4, Sort by order · gradasi biru → oranye → merah, label data aktif · Reference line di Netral (7.597) · catatan kaki: "La Niña kuat: only 3 months observed — too thin to conclude"

gambar-sac/h07-fase-enso.png

Same signal, seasonality removed

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Heat map — Lag correlation - anomaly

Bukti "bukan musim". Cadangan kalau juri menekan: *Province consistency IDN* — bar horizontal, Measure ratio_nino_vs_other, reference line di 1,0; 33 dari 34 batang ada di kanan garis.

gambar-sac/h07-anomali.png

ANALYSIS 4 — WHERE AND WHEN

The alarm must be set province by province

- **Indonesia:** 20 of 35 provinces peak in **January**, 8 in February — but **Bali peaks in May** and North Kalimantan in December
- **Thailand:** 38 of 77 provinces peak in **July**, 16 in August
- A single national alert date would be wrong for a third of Indonesia

Indonesia — share of annual cases by month

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Heat map — *Risk calendar IDN*

Dim baris province (urut menurut bulan puncak, *bukan* alfabetis) ·
Dim kolom month · Measure share_percent · skala sekuensial putih
→ merah · baris Bali diberi garis tebal

gambar-sac/h08-kalender-idn.png

Thailand — share of annual cases by month

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Heat map — *Risk calendar THA*

Setelan identik. Kontras dua panel inilah pesannya: satu penyakit, dua kalender.

gambar-sac/h08-kalender-tha.png

ANALYSIS 5 — WHO GETS HIT HARDEST

The same alarm, aimed at the most exposed places

- Highest incidence 2018–2020: **Bali 134**, North Kalimantan 115, East Kalimantan 112 per 100k — **14× the lowest province** (Papua, 9)
- Indonesia has the **lowest piped-water access in ASEAN: 21.8%** (JMP 2024) — households store water, and stored water breeds *Aedes*
- **12.9 million people in Bangkok** and 1.7 million in Jakarta live in the 1-in-100-year flood zone
- We use this to **target** the alarm, not to explain it (see p. 14)

Incidence per 100k, 2018–2020

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Bar chart horizontal — *Province incidence 18-20*

Measure mean · Dim province,urut menu-run, ambil 10 teratas + 3 terbawah · 3 teratas merah

gambar-sac/h09-insidens.png

Piped water access, ASEAN

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Bar chart — *WASH by country*

Measure piped_water_% · Dim country,urut menaik · Indonesia paling kiri, oranye

gambar-sac/h09-wash.png

Flood-zone population

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Bar chart — *City flood vulnerability*

Measure exposed_2020 · Dim city, 8 kota teratas

gambar-sac/h09-banjir.png

SYNTHESIS

From signal to schedule: a risk calendar every health office can read

- Each province gets three states — **Peak** · **Watch** · **Normal** — from its own case history
- The El Niño index sets **how severe** the coming peak will be; the calendar sets **when** it arrives
- Real lead times, every El Niño episode since 2009: **0–3 months** on observed ONI — and longer on NOAA/BMKG *forecast* ONI

Risk calendar, six priority provinces

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Heat map — *Risk calendar IDN*

Filter 6 provinsi: Bali, Kalimantan Utara, Kalimantan Timur, Gorontalo, DKI Jakarta, Jawa Barat · Measure `status_code` · palet 3 warna: 0 abu (Normal), 1 kuning (Watch), 2 merah (Peak)

gambar-sac/h10-kalender-prioritas.png

Cases vs the P75 outbreak threshold

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Time series — *Monthly series IDN-THA*

Filter Indonesia · Line `cases` · Reference line `threshold_p75` (11.932) · warna titik/latar menurut `alarm_el_nino`

gambar-sac/h10-deret-alarm.png

THE SOLUTION

Two conditions. No model. No code.

“If the month is Jan–Apr AND ONI four months ago was $\geq 0.5 \rightarrow$ raise alert”

11.6%

of months fire the alert

100%

precision — every alert was a top-quartile dengue month

4×

lift over the base rate

- Both inputs are public and free: NOAA publishes ONI monthly, the calendar comes from the office's own records
- Runs in a spreadsheet, a WhatsApp broadcast, or this SAC dashboard — competitors promise machine learning; we promise something a provincial health office can start next week

When the rule fired, 2010–2024

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Time series — *Monthly series IDN-THA*

Filter Indonesia · Line cases · warna menurut alarm_rule_d · Reference line threshold_p75

gambar-sac/h11-deret-aturan.png

Precision vs alert frequency, four rules

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Bar chart — *Trigger rules - evaluation*

Measures precision_% dan alarm_coverage_% ·
Dim rule · baris D merah, sisanya abu — aturan
D menyala paling jarang tapi paling tepat

gambar-sac/h11-aturan.png

VIABILITY

We tried to break our own rule. It held.

- **Out-of-sample test:** rule built on 2010–2016, tested blind on 2017–2024 → **precision still 100%**, sensitivity 44%, lift 3.9
- Against the **WHO/PAHO endemic-channel** definition — which removes seasonality by construction — specificity **91%**, lift 2.5, and *all* of the value comes from the El Niño condition, not the season
- Published dengue EWS report PPV 43–86% (17-system review); we reach a comparable PPV **with no statistical model and monthly data**
- Replicates in Thailand with its own best predictor (temperature anomaly, lag 2): lift 2.7

ADOPTION IN THREE PHASES

1 Bali, N. & E. Kalimantan pilot → 2 all 38 provinces
→ 3 ASEAN peers

Cost: open data, monthly refresh, existing staff.

Train vs blind test, Indonesia and Thailand

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Bar chart — Trigger rules - out of sample

Measures precision_%, sensitivity_% · Dim period (latih vs Uji), dikelompokkan per country

gambar-sac/h12-luar-sampel.png

How we compare to published EWS

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Table widget — EWS benchmark

Tampilkan 5 baris apa adanya, *termasuk* kolom sumber — kejujuran pembandingan inilah yang dinilai

gambar-sac/h12-tolok-ukur.png

IMPACT — INDONESIA ALONE

Acting 4–8 weeks earlier is worth US\$1.5–3.7 million a year

4,656

cases/year prevented in Indonesia, under the most conservative scenario

Avoided hospitalisation cost by scenario

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Bar chart bertingkat — *Monetary impact*

Measures hospitalisation low (US\$316) dan hospitalisation high (US\$791) · Dim scenario · tambahkan Numeric Point (KPI tile) besar: 4,656 cases/year prevented

gambar-sac/h13-dampak.png

Lead time, every El Niño episode since 2009

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Table — *El Nino episodes - lead time*

5 baris: episode, alarm_fires, cases_cross_p75, lead_time_months

gambar-sac/h13-tenggang.png

- Indonesia averages **104,753 cases/year** (complete years 2015–2023); **44% fall in Jan–Apr**
- Prevent just **10%** of peak-season cases (an assumption, not a finding) = 4,656 cases/year
- At Indonesia's published hospitalisation cost of US\$316–791/case → **US\$1.5–3.7 M (Rp 24–59 billion)/year**
- For scale: dengue costs Indonesia **US\$381–681 M/year** (Nadjib 2019; Wilastonegoro 2020)
- Lead time is real: in the 2023–24 episode the alarm fired **Sep 2023**; cases crossed the peak threshold **Dec 2023**

LIMITS & DATA ETHICS

What we tested and threw away

- **We falsified our own innovation angle.** “Drought $\times$ water access” looked promising ($r = -0.31$) but collapsed after controlling for El Niño (partial $r = -0.15$) and showed no moderation by piped access ($\rho = +0.34$, $p = 0.46$, $n = 7$). **We dropped the claim.**
- **We do not claim a 2023–24 record** — our panel is incomplete for Vietnam and the Philippines after 2022.
- **Incidence only for 2018–2020**, the only years with official BPS denominators. Elsewhere we use raw counts and never rank regions by them.
- **Flood and disaster records look correlated with dengue ($r = +0.66$) — until you divide by population ($r = -0.14$).** We report the null.
- Correlation is not causation: this is an **early-warning** claim, with mechanism cited from the literature.

No relationship ($\rho = +0.34$, $p = 0.46$, $n = 7$)

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Scatterplot — *Drought $\times$ water access*

X piped_water_% · Y r_drought · ukuran bubble nino_multiplier · label country · **sengaja tanpa garis tren** — judul grafik menyatakan hasil nolnya

gambar-sac/h14-kekeringan.png

HALAMAN CADANGAN

Kuota resmi 15 halaman — satu halaman masih kosong

Halaman 1–14 sudah memakai 14 dari 15 halaman yang diizinkan (halaman Referensi tidak dihitung). **Satu halaman masih tersisa.**

Dari 15 dek finalis ASEAN DSE 2024–2025 yang dianalisa, **seluruhnya** menutup bagian solusi dengan sebuah **produk bernama beserta mockup antarmuka** — CircularEat, Agrigrow, MentaLink, GreenLoop, Nurture Hub. Panitia sendiri menyatakan tim “encouraged to create an app or technological prototype” untuk memperkuat pencalonan di National Finals.

Storyboard ini berhenti pada *aturan*, bukan *produk*. Halaman ini adalah tempatnya: satu nama, satu mockup dashboard SAC atau aplikasi, dan siapa yang menekan tombolnya.

Isi halaman ini, atau hapus seluruh blok halaman ini di berkas .tex — dek tetap sah dengan 14 halaman.

REFERENCES

Data sources and literature

DATA

OpenDengue V1.3 (doi:10.6084/m9.figshare.24259573.v4) · World Bank Climate Change Knowledge Portal (ERA5 0.25°) · NOAA CPC Oceanic Niño Index · WHO/UNICEF Joint Monitoring Programme 2025 · BPS Indonesia (WebAPI) · Global Human Settlement Layer (JRC) · SIPSN KLHK · BNPB.

LITERATURE

Tian et al. 2025 *Nat Commun* · Djaafara et al. 2026 *PLOS NTD* · Sutriyawan et al. 2025 · Lowe et al. 2021 *Lancet Planet Health* · Gibb et al. 2023 *Nat Commun* · Baharom et al. 2022 · Cárdenas et al. 2021 · Schlesinger et al. 2024 · Hussain-Alkhateeb et al. 2018 (WHO-TDR EWARS) · Nadjib et al. 2019 *PLoS NTD* · Wilastonegoro et al. 2020 *AJTMH* · Sasmono et al. 2026 *IJID Regions* · Childs et al. 2025 *PNAS*.

TODO sebelum submit: ganti seluruh pranala consensus.app di `analisis/keluaran/referensi_per_klaim.md` dengan DOI jurnal, lalu lengkapi sitasi di halaman ini menjadi format penuh (penulis, tahun, judul, jurnal, DOI). Halaman referensi tidak dihitung dalam batas 15 halaman — boleh lebih dari satu halaman.